Benchmarking Machine Learning Models for Road-Level Flood Prediction and Flood-Aware Routing: A Case Study of Da Nang, Vietnam

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***Abstract*—Flood studies in Vietnamese cities map susceptibility over areas rather than over the roads that vehicles use, and model comparisons usually stop at classification scores. This paper benchmarks six machine learning models from five families—logistic regression, support vector machine, random forest, XGBoost, multilayer perceptron, and the graph neural network GraphSAGE—both for road-level flood prediction and for the flood-aware routing decisions built on their predictions, using the extreme rainfall of 14 October 2022 in Hai Chau, the central district of Da Nang, Vietnam, as a case study. A flood inventory of [need results] OpenStreetMap road segments was built from verified multi-source flood reports, and each segment was described by terrain, hydrological, imperviousness, and network features. Under spatial block cross-validation, [need results: best classifier] achieved the highest area under the precision–recall curve ([need results]), while random cross-validation overstated AUC-ROC by [need results: range]. The out-of-fold probabilities of each model were then used as routing costs and compared with shortest-path and oracle baselines. [need results: best routing model] reduced exposure to flooded segments by [need results]% at a median travel-time increase of [need results]%, and [need results: whether the best classifier was also the best for routing]. Shapley additive explanations identified [need results: top features] as the main flood drivers.**

***Index Terms*—benchmarking, explainable artificial intelligence, flood-aware routing, graph neural networks, machine learning, road networks, spatial cross-validation, urban flooding**

I. Introduction

Urban flooding disrupts road transport long before it damages buildings. A road becomes unsafe for most vehicles at about 30 cm of standing water [**?**], and the disruption spreads beyond the flooded links. During Hurricane Harvey, flooding of only 1.3% of the roads in Houston increased network-wide travel times by 8% [**?**]. For drivers and traffic managers, the useful question during heavy rain is therefore which road segments become impassable, not which areas are susceptible.

Da Nang, Vietnam, illustrates this problem. On the night of 14 October 2022, the city received close to 600 mm of rain; parts of the city stood about 1 m deep, and the Dien Bien Phu underpass flooded completely [**?**], [**?**]. Hai Chau, the dense central district along the Han River, is classed among the most flood-susceptible areas of the city [**?**].

Research on flooding in Da Nang has grown quickly. Recent studies map flood susceptibility with machine learning [**?**], [**?**], simulate the sewer network for the 2022 event [**?**], and map inundation from Sentinel-1 radar imagery [**?**]. All of them report results for areas, grid cells, or sewer nodes, and even advanced inundation models cannot reliably resolve water levels on individual roads [**?**]. Road-level prediction has been demonstrated elsewhere: Yuan et al. [**?**] predicted flooded roads in Houston from crowdsourced and traffic data. However, their models were validated with random splits, and their predictions were not tested for routing. Moreover, model comparisons in flood studies cover mostly tree ensembles [**?**], [**?**], [**?**], [**?**], so linear, kernel, neural, and graph-based models are rarely tested on the same road-level data. Finally, flood models are compared by classification scores alone, so it remains unknown whether the model with the best score also leads to the best routing decisions.

This paper addresses these gaps with a benchmark of six machine learning models from five families for road-level flood prediction and flood-aware routing in Hai Chau during the 2022 Da Nang flood. Its contributions are as follows:

1) A segment-level flood inventory for the 14 October 2022 event, built with a documented multi-source procedure for collecting, verifying, and assigning flood labels.
2) A benchmark of linear, kernel, tree-ensemble, neural, and graph models under one leakage-aware protocol, covering spatial block cross-validation, precision–recall and calibration metrics, statistical tests, and SHAP interpretation.
3) A downstream benchmark in which the out-of-fold flood probabilities of every model are used for flood-aware routing and compared with shortest-path and oracle baselines.
4) An analysis of whether classification scores agree with routing performance when ranking the models.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the study area, the data, and the construction of flood labels, and Section IV presents the benchmark design. Section V reports the results, and Section VI discusses the findings and concludes the paper.

II. Related Work

A. Flood Studies in Da Nang

Nguyen et al. [**?**] compared five tree ensembles across the city (AUC 0.983–0.989) and placed Hai Chau in the

high and very-high susceptibility classes. Hoang et al. [?] trained XGBoost on 1,000 points drawn from 2022–2023 media reports and reached an accuracy of 0.93. Both studies used random train–test splits, which overstate accuracy when nearby samples are correlated [?]. Doan et al. [?] applied MIKE Urban to the 2022 rainfall and found that riverside manholes flooded when high river levels blocked the outfalls. Nguyen and Quach [?] mapped about 10,505 ha of flooding from Sentinel-1 imagery in October 2022, mostly outside the dense urban core.

B. Road-Level Flood Prediction and Sensing

Ha et al. [?] mapped flash-flood susceptibility along National Highway 6 in Hoa Binh (AUC 0.991), a rural Vietnamese precedent. Pregnolato et al. [?] related flood depth to vehicle speed and set 30 cm as the depth at which a road becomes impassable. Sadler et al. [?] modeled street flood severity from crowdsourced reports with random forest. Yuan et al. [?] are closest to this work. They predicted flooded roads in Harris County, Texas, from topographic, hydrologic, and precipitation features, labeling roads from Waze reports and INRIX traffic data. Random forest reached AUC values of 0.86 for Hurricane Harvey and 0.79 for Tropical Storm Imelda, but under random five-fold cross-validation. Pan et al. [?] coupled road flood states and traffic speeds in a two-layer Bayesian network for New York City expressways to decide where to place road flood sensors. Their work observes flooding on segments, whereas this paper predicts which segments are prone to it. Graph neural networks such as GraphSAGE [?] can use the connectivity between segments directly, but they have not been compared with feature-based models for road-level flood prediction. [To do: summarize [?] and [?].]

C. Routing Under Disruption

Le et al. [?] optimized urban bus routes with multi-objective deep reinforcement learning, without hazard costs. Linking segment-level flood probabilities to routing connects these two lines of work.

III. Study Area and Data

A. Study Area and Historical Flood Event

The study covers the former Hai Chau district of Da Nang, with boundaries as they stood in October 2022, on the west bank of the Han River. The district contains the commercial core of the city, a dense street grid, and the Dien Bien Phu underpass, and it is classed among the most flood-susceptible areas of the city [?].

The historical event studied is the extreme rainfall of 14–15 October 2022. Da Nang received close to 600 mm of rain overnight, parts of the city stood about 1 m deep, and the Dien Bien Phu underpass flooded completely [?], [?]. Manholes along the Han River also overflowed, because high river levels blocked the drainage outfalls [?]. [need results: station rainfall totals and peak hourly intensity, 14–15 October 2022.] Fig. 1 shows the study area.

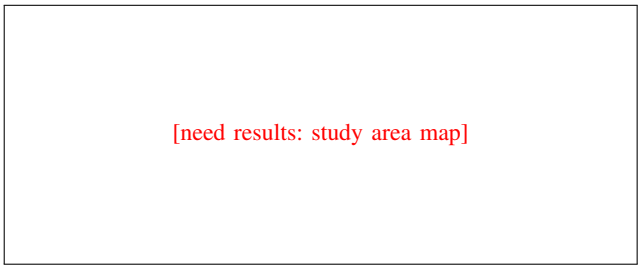


Fig. 1. Study area, road network, and observed flooded segments in Hai Chau.

B. Road Network and Flood Labels

The drivable network was obtained from OpenStreetMap with OSMnx [?] and split at intersections and at a maximum length of 100 m, giving [need results] segments. Flood labels are then built in four steps.

1) Collection: Flood observations for 14–15 October 2022 are gathered from [need results: primary 2022 flood records, e.g., city agency reports], geotagged news reports, and social media posts. Each observation keeps its location, time, source, and reported water depth when available.

2) Geocoding and snapping: Street names and addresses are geocoded, and each observation is snapped to the nearest segment within 30 m. Observations farther than 30 m from any segment are discarded.

3) Verification: An observation is kept only if a second source or visual evidence, such as a photo or video, confirms it; [need results: share of observations verified]. Sentinel-1 flood extents [?] serve only as a cross-check, because radar underdetects flooding in narrow streets.

4) Label assignment: A segment is labeled flooded ($y = 1$) if at least one verified observation falls on it. Segments without an observation are labeled non-flooded ($y = 0$). Segments within [need results: buffer distance] of a flooded segment but without their own report are treated as uncertain and excluded, because unreported flooding is likely there; a sensitivity run that keeps them as non-flooded is also reported. When depth is reported, segments are further classed as passable (< 10 cm), flooded but passable (10–30 cm), or impassable (≥ 30 cm), following [?].

The final inventory contains [need results] flooded, [need results] non-flooded, and [need results] excluded segments.

C. Features

Each segment is described by features computed within a 25–50 m buffer. Terrain features are the minimum and mean elevation, slope, and topographic position index from a [need results: DEM source and resolution] digital elevation model (DEM). Hydrological features are the distances to the Han River and to the nearest canal, retention pond, and drainage outfall. Imperviousness is the mean normalized difference built-up index (NDBI) from Sentinel-2 imagery. Network features are road class, length, node degree, and betweenness centrality.

Fig. 2. Benchmark workflow from flood labels to the downstream routing comparison.

IV. Benchmark Design

Fig. 2 summarizes the benchmark. The six models are compared at two levels: how well they classify flooded segments, and how well their predictions support routing.

A. Models and Common Protocol

For each segment e_i with feature vector $\mathbf{x}_i$, a model estimates the probability of flooding:

$$p_i = P(y_i = 1 \mid \mathbf{x}_i). \tag{1}$$

For the graph model, the input also includes the features of neighboring segments in the road network. Six models from five families are compared:

- **Linear:** logistic regression (LR), a transparent baseline.
- **Kernel:** support vector machine (SVM) with a radial basis function kernel [**?**], with probabilities from Platt scaling.
- **Tree ensembles:** random forest (RF) [**?**], based on bagging, and XGBoost [**?**], based on gradient boosting.
- **Neural:** multilayer perceptron (MLP) with [need results: architecture].
- **Graph:** GraphSAGE [**?**], which aggregates the features of adjacent segments on the line graph of the road network.

All models receive the same features, the same folds, and the same random seeds. Class weights handle class imbalance, features are standardized for LR, SVM, MLP, and GraphSAGE, and hyperparameters are tuned by [need results: search method] inside the training folds only. [need results: confirm the final model settings with the experiment team.]

B. Classification Benchmark

Neighboring segments share terrain and drainage, so random splits leak information between training and test sets [**?**]. The district is therefore divided into square blocks of [need results: block size] m, which are assigned to five folds and held out in turn. For GraphSAGE, held-out segments stay in the graph, but their labels are hidden during training. Random five-fold cross-validation is also run, and the optimism gap of model m is

$$\Delta_m = \mathrm{AUC}_m^{\mathrm{random}} - \mathrm{AUC}_m^{\mathrm{spatial}}. \tag{2}$$

Flooded segments are the minority class, so the primary metric is the area under the precision–recall curve (AUC-PR). The area under the receiver operating characteristic curve (AUC-ROC), F1 score, and recall on the flooded class are also reported. Because the probabilities become routing costs, calibration is measured with the Brier score [**?**]. Differences across folds are tested with the Friedman test and pairwise Wilcoxon signed-rank tests [**?**]. SHAP values [**?**] are computed for every model, using TreeSHAP for RF and XGBoost and KernelSHAP for the other models, to compare the flood drivers they learn.

[need results: road risk map of the best model]

Fig. 3. Out-of-fold flood probability per road segment from the best model, in three risk classes.

C. Downstream Routing Benchmark

For each model, every segment receives the probability predicted in the fold where it was held out. These out-of-fold probabilities form a district-wide risk map, so no model routes on labels it was trained on. The probabilities are converted into edge costs:

$$C_i = t_i\,(1 + \lambda\, p_i), \tag{3}$$

where t_i is the free-flow travel time of segment e_i and λ controls risk aversion. For [need results: number of pairs] random origin–destination (OD) pairs, routes are computed with Dijkstra's algorithm for each model and for two baselines: shortest time ($p_i = 0$), which ignores flooding, and an oracle ($p_i = y_i$), which knows the observed flood labels. All strategies use the same OD pairs, λ, and network. Each route is scored by the number and length of observed flooded segments it uses, its travel-time increase over the shortest route, and whether a flood-free route exists. Finally, the model rankings by classification and by routing metrics are compared to test whether they agree.

V. Results

A. Flood Inventory

Of the [need results] segments, [need results] ([need results]%) were observed flooded and [need results] were excluded as uncertain. The flooded segments cluster [need results: spatial pattern, e.g., along the Han River or near the underpass].

B. Model Performance

Table I compares the six models. [need results: name the best model by AUC-PR and recall, the Friedman/Wilcoxon results, and the gap between spatial and random cross-validation.] [need results: effect of keeping uncertain segments as non-flooded (sensitivity run).]

C. Risk Map and Feature Effects

Fig. 3 maps the out-of-fold flood probability of each segment for the best model. [need results]% of the segments fall in the high-risk class, concentrated in [need results: area]. Fig. 4 summarizes the SHAP values. [need results: top three features, direction of their effects, and whether the six models agree on them.]

TABLE I
CLASSIFICATION BENCHMARK UNDER SPATIAL BLOCK CROSS-VALIDATION (MEAN ± STANDARD DEVIATION OVER FIVE FOLDS)

		Spatial CV					Random CV
Model	Family	AUC-PR	AUC-ROC	F1	Recall (flooded)	Brier score ↓	AUC-ROC
Logistic regression	Linear	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
SVM (RBF)	Kernel	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
Random forest	Tree ensemble	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
XGBoost	Tree ensemble	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
MLP	Neural	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]
GraphSAGE	Graph	[NR]	[NR]	[NR]	[NR]	[NR]	[NR]

Best value per column in bold. ↓: lower is better. [NR]: need results.

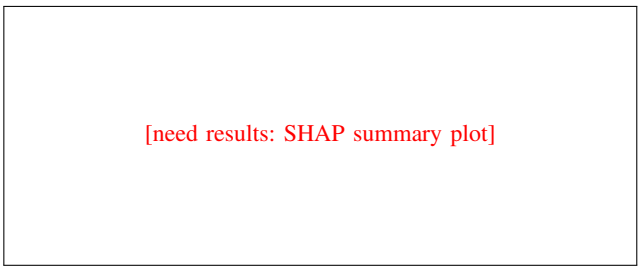


Fig. 4. SHAP summary plot of the best model.

TABLE II
ROUTING RESULTS OVER [NR] OD PAIRS (λ = [NR])

Strategy	Flooded seg. per trip	Flooded length (m)	Time increase (median, %)	No safe route (%)
Shortest time	[NR]	[NR]	0	–
LR	[NR]	[NR]	[NR]	[NR]
SVM (RBF)	[NR]	[NR]	[NR]	[NR]
Random forest	[NR]	[NR]	[NR]	[NR]
XGBoost	[NR]	[NR]	[NR]	[NR]
MLP	[NR]	[NR]	[NR]	[NR]
GraphSAGE	[NR]	[NR]	[NR]	[NR]
Oracle (observed)	0	0	[NR]	[NR]

D. Downstream Routing

Table II compares the routing results of the six models with the two baselines. [need results: which model gives the fewest flooded segments, how close it comes to the oracle, and at what travel-time cost.] [need results: whether the ranking by routing matches the ranking by AUC-PR in Table I, and which classification metric best explains the routing ranking, e.g., recall or Brier score.]

VI. DISCUSSION AND CONCLUSION

[need results: interpret which mechanisms the models recover, e.g., tide-locked outfalls near the river, as reported in [**?**].] [need results: explain why the best classifier was or was not the best for routing, e.g., missed flooded segments (low recall) or poorly calibrated probabilities change the routes more than AUC suggests.] For practitioners, the benchmark indicates that models for flood-aware routing should be selected by their routing outcomes, or at least by recall and calibration, not by AUC alone.

Four limitations apply. First, the study uses one event, so it identifies flood-prone segments rather than forecasting floods for any storm. Second, unreported flooding outside the excluded buffer is still labeled as non-flooded, which lowers the measured recall. Third, rainfall varies little across a single district, so it cannot separate segments within one event. Fourth, a coarse DEM may miss street-scale depressions.

In conclusion, this paper benchmarked six machine learning models from five families for road-level flood prediction and flood-aware routing during the 2022 Da Nang flood. [need results: best classifier] reached the highest AUC-PR of [need results] under spatial cross-validation, and [need results: best routing model] reduced flood exposure by [need results]% for a [need results]% increase in travel time. Future work will add further events, timestamped depth records for short-term prediction, and risk costs in reinforcement-learning routing [**?**]. The risk maps could also provide prior flood probabilities for placing road sensors or cameras in Da Nang, following [**?**].

REFERENCES